CS3235 — LLM / Multimodal AI Projects

Mini-Project 3

PatraAI

Indian Document Understanding Tool

Final Project Report

Extracting fields, summarising, translating and answering questions about Indian government documents

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Course	CS3235 — LLM / Multimodal AI Projects
Project	3. Indian Document Understanding Tool
GitHub repository	https://github.com/pavankumarkn04/PatraAI
Date	November 2026

Abstract

Government notices, circulars, scholarship letters, certificates and application forms in India are often scanned, long and written in a language the reader does not know well. PatraAI is a web application that helps people understand such documents. A user uploads a PDF, Word file or image; PatraAI extracts the key details (issuing office, date, reference number, fees, deadlines, eligibility, required documents) with the page each came from, writes a plain-language summary, translates the results into Kannada, Hindi, Tamil, Telugu or Malayalam with AI4Bharat IndicTrans2, and answers questions in any of these six languages using retrieval-augmented generation.

The project sheet specifies BharatGen Patram and AI4Bharat IndicTrans2. IndicTrans2 is used for all translation. Patram needs more memory than the project laptop has and could not be run, so a smaller multimodal model of the same kind (qwen3.5:4b, served by Ollama) does the reading and answering; the application can switch to Patram through one setting when a GPU server is available. Everything runs on a laptop CPU without a GPU.

The application was evaluated on 45 hand-checked test cases from 18 real public documents in six languages and nine formats. It reached 64.4% field-extraction accuracy and 66.7% question-answering accuracy. Most errors came from OCR of scanned Kannada and Malayalam pages (10 of 31 wrong checks) and from long documents where the right page was not retrieved (10). A failure analysis traces every wrong result to its root cause, and 98 automated tests check the software.

Table 1**:** Key results

What was measured	Result
Test cases	45 cases over 18 real public documents, 6 languages
Field-extraction accuracy	64.4% (29 of 45 fields correct)
Question-answering (QA) accuracy	66.7% (30 of 45 answers correct)
Answer cited the correct page	66.7%
Automated software tests	98 tests, all passing
Speed-up from optimisation	analysis 23% less time, translation 50% less time, same accuracy

Keywords: document understanding, OCR, multimodal LLM, retrieval-augmented generation, IndicTrans2, Indian languages

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Rubric and Requirements Coverage

The project sheet assesses six criteria and requires 30–50 representative test cases, a failure analysis, a usable frontend and a GitHub repository. The tables below show where each is covered in this report.

Table 2: Where each rubric criterion is covered

Rubric criterion	Chapter	What is covered
Problem definition & relevance	1	Who needs the tool and why Indian documents are hard to read automatically
Model / dataset selection	2	Patram and IndicTrans2 (as specified), why Patram could not run on the laptop, the model used instead, and the 45-case test dataset
Working end-to-end application	3	How the pipeline works, with screenshots of the running Streamlit application
GenAI / RAG / agentic integration	4	LLM extraction, retrieval-augmented question answering (RAG), a self-checking language loop and hallucination checks
Quantitative testing & error analysis	5	Accuracy on 45 test cases, failure analysis with root causes, speed measurements, 98 automated tests
Demo, documentation & repository	6	GitHub repository, how to run the application, demo script

Table 3: Project-sheet requirements

Requirement	Status	Section
30–50 representative test cases	Done: 45 cases	2.4, 5, App. A
Failure analysis	Done	5.4
Usable frontend	Done: Streamlit web application	3
GitHub repository	Done	6.1
Check model availability, licence, languages and GPU needs	Done for Patram and IndicTrans2	2.1, 2.2
BharatGen Patram	Supported, but not run (needs a GPU server)	2.1
AI4Bharat IndicTrans2	Used for all translation	2.2

Terms used in this report

Table 4: Terms used in this report

Term	Meaning
LLM	Large language model: an AI model that reads and writes text. PatraAI uses one that can also read images.
OCR	Optical character recognition: reading the text out of a scanned page or photo.
Field extraction	Pulling specific details (date, fee, issuing office, ...) out of a document.
RAG	Retrieval-augmented generation: find the most relevant parts of the document first, then let the LLM answer only from those parts.
Searchable / scanned PDF	A searchable PDF contains real text; a scanned PDF only contains pictures of pages, so it needs OCR.
Legacy font	Older Kannada fonts (Nudi, Baraha) whose text comes out as gibberish when copied, so the page must be read with OCR.

1. Problem Definition and Relevance

In short: Important government documents in India are often scanned, long, and written in a language the reader does not know well. PatraAI helps people find what matters in them, in their own language.

Government circulars, scholarship notices, scheme orders, admission brochures, certificates and application forms decide who is eligible for something, which documents to submit and by when. They are issued in English and in regional languages, often as scanned PDFs, and are frequently long.

Example. A parent in Karnataka who reads Kannada but not English receives a 7-page English scholarship letter. They need to know: Is my child eligible? How much is the amount? What is the deadline? Which office issued it? PatraAI answers these in Kannada and shows the page each answer came from, so it can be checked.

PatraAI works on any document; nothing is tied to one template or department. The main design goal is trust: a value that is not in the document is not shown, and every value says where it came from.

1.1 Why this is hard

- **Many scripts and old fonts.** Six scripts are supported. Many Kannada PDFs use legacy fonts whose text cannot be copied correctly, so PatraAI must notice this and use OCR instead.
- **Scanned pages.** Pages that are only pictures need OCR before anything else can happen.
- **Small model, no GPU.** On a laptop CPU the model writes only about 9 word pieces (tokens) per second, so prompts must be short, and its output must be checked for invented numbers and wrong page numbers.
- **Cross-language questions.** A question in Kannada about an English document must be answered in Kannada, using English evidence.

2. Model and Dataset Selection

In short: The project sheet names BharatGen Patram and AI4Bharat IndicTrans2. IndicTrans2 is used for all translation. Patram is too large for the laptop, so a smaller model of the same kind (qwen3.5:4b) does the reading and answering, and the application can switch to Patram when a GPU server is available.

Table 5: Models and their roles

Task	Model	Runs on	Used in this report?
Read the document, extract fields, summarise, answer questions	BharatGen Patram 7B (specified)	GPU server	No: supported, but no GPU was available
Same tasks (actually used)	qwen3.5:4b via Ollama (reads text and images)	Laptop CPU	Yes, for every result
Translate English into 5 Indian languages	AI4Bharat IndicTrans2 (200M, distilled)	Laptop CPU	Yes
Backup if a model reply fails	gemma4:e4b	Laptop CPU	Only as a retry (used for 1 scanned page)

2.1 BharatGen Patram: why it was not run

Before choosing a model, the project sheet asks to check its availability, licence, languages and hardware needs. For Patram-7B-Instruct this check showed:

- its weights alone need about 15.3 GB of memory, but the laptop has 15.2 GB of RAM in total and no GPU;
- Hugging Face offers no hosted service that runs it, and there is no smaller (quantised) version;
- it uses a custom model design that Ollama cannot run.

So Patram cannot run on the project laptop. The application is still built for it: pointing one setting (HF_CHAT_COMPLETIONS_URL) at a Patram server sends all reading, OCR and question answering to Patram. The application never pretends Patram was used: its Processing Details panel names the model that actually answered and marks Patram as "Not configured". This gap is listed again in Chapter 7.

2.2 AI4Bharat IndicTrans2

IndicTrans2 translates the summary, the extracted details and the answers from English into Kannada, Hindi, Tamil, Telugu and Malayalam on the laptop CPU. Numbers, dates, e-mail

addresses and reference codes are protected so translation never changes them; if a protected value goes missing, the sentence is translated again.

2.3 Why qwen3.5:4b

- It fits in the laptop's free memory (3.4 GB). The larger gemma4:e4b (9.6 GB) did not fit and slowed to a crawl.
- It reads both text and page images, so one model handles OCR and analysis without reloading.
- Ollama can force its reply into a fixed JSON structure, which makes field extraction reliable.

2.4 Evaluation dataset

The test set has 45 test cases over 18 documents, all taken from real public sources (government and board websites). 42 cases use the original files; 3 cases use files derived from them (pages 1–3 of a 22-page Kerala gazette, and page 1 of a Tamil Nadu letter saved as a JPG image). Every expected answer was read by hand from the original document; none came from the model being tested. Each test case has:

- one expected field value (for example the date or reference number) and the page it is on;
- one question, its correct answer and the page it is on.

Table 6: Coverage of the test dataset

Coverage	Number of test cases
Languages	English 25, Hindi 7, Kannada 6, Tamil 3, Malayalam 2, English + Punjabi 2
Formats	Searchable PDF 18; Scanned PDF 7; Mixed PDF (searchable + scanned pages) 6; Scanned PDF with OCR text layer 3; Legacy-font PDF (needs OCR) 3; Long PDF (83 pages, mostly searchable) 3; Mixed PDF (searchable + scanned page) 2; Word (DOCX) 2; Image (JPG) 1
Document types	Government circular 6; Vacancy circular 4; Scholarship notification 4; Admission brochure 3; Press release / notice 3; Official circular / letter 3; Government scheme order 3; Public policy document 3; Gazette notification 2; Notification with schedule table 2; Government order 2; Official proceedings / letter 2; Government certificate (prescribed format) 2; Application form + certificate format 2; Application form 2; Educational document 2

Telugu: no public Telugu government document with real (Unicode) text could be found, so none is in the test set. Telugu questions and translation are supported and covered by automated tests, but Telugu document understanding is not measured.

3. Working End-to-End Application

In short: PatraAI is a one-page Streamlit web application. A document goes through ten steps, from upload to results. It runs fully on the user's computer; no document is sent to an outside service.

3.1 How it works

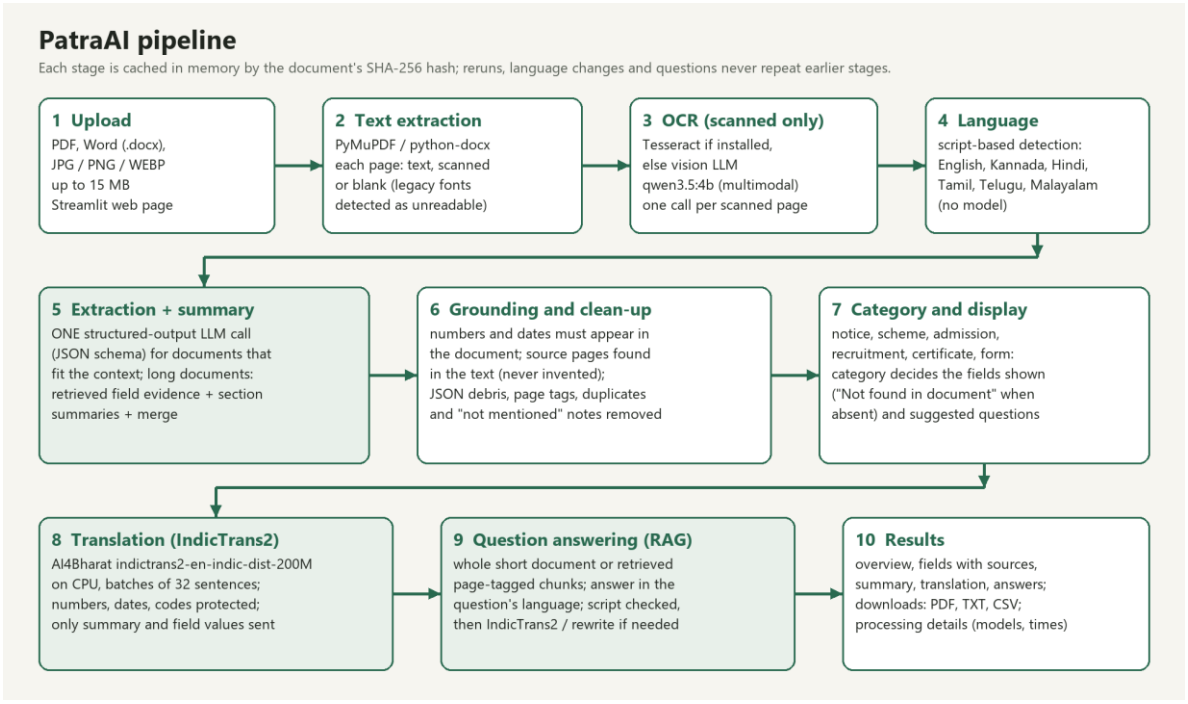


Figure 1: The PatraAI pipeline. Shaded steps use AI models; the other steps are ordinary code.

In simple terms: the application reads the text of each page (using OCR only for scanned pages), works out the language, asks the model to fill in a fixed form of fields and a summary, checks the model's output against the document, and then translates and answers questions on request. Every step's result is kept in memory, so changing the language or asking another question never repeats earlier work.

3.2 Screenshots of the running application

Figure 2 shows the upload panel. Figures 3 to 6 show the application analysing the CBSE press release on the three-language policy (4 pages).

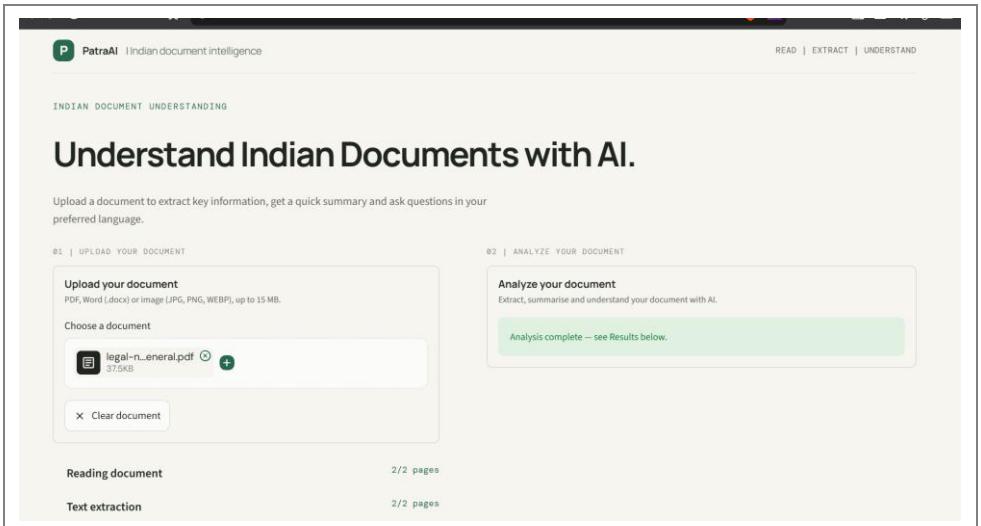


Figure 2**:** Upload panel (PDF, Word or image, up to 15 MB). Reading and text-extraction progress is shown page by page.

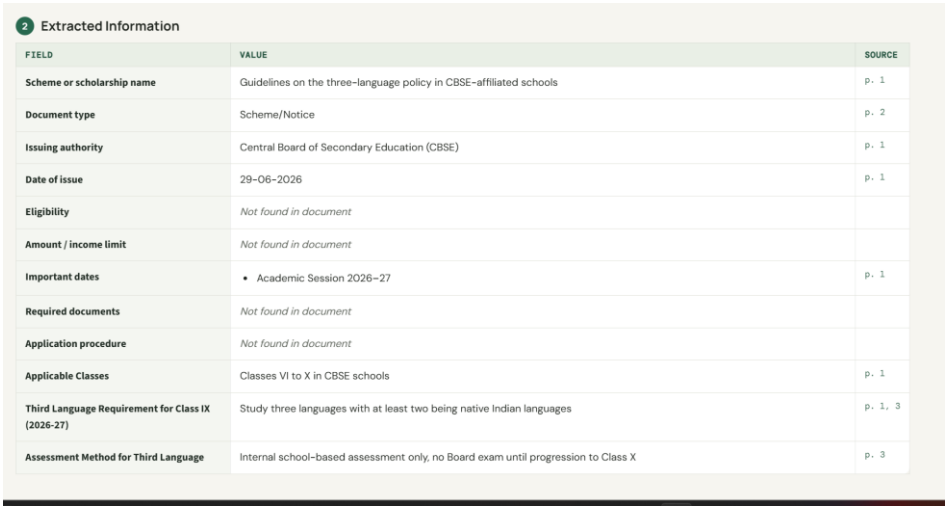


Figure 3**:** Extracted information. Every value shows its source page; missing fields say "Not found in document" instead of guessing.

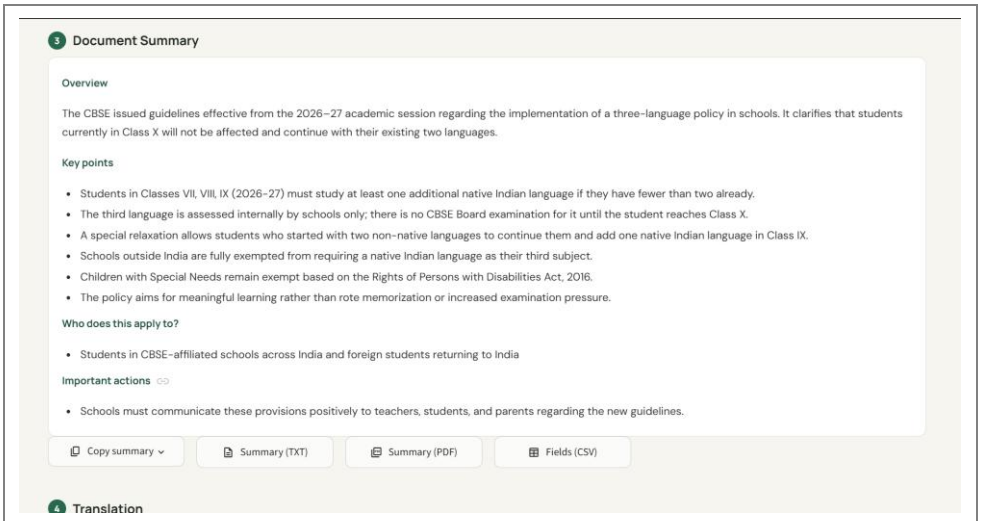


Figure 4: Short summary, with downloads (TXT, PDF, CSV).

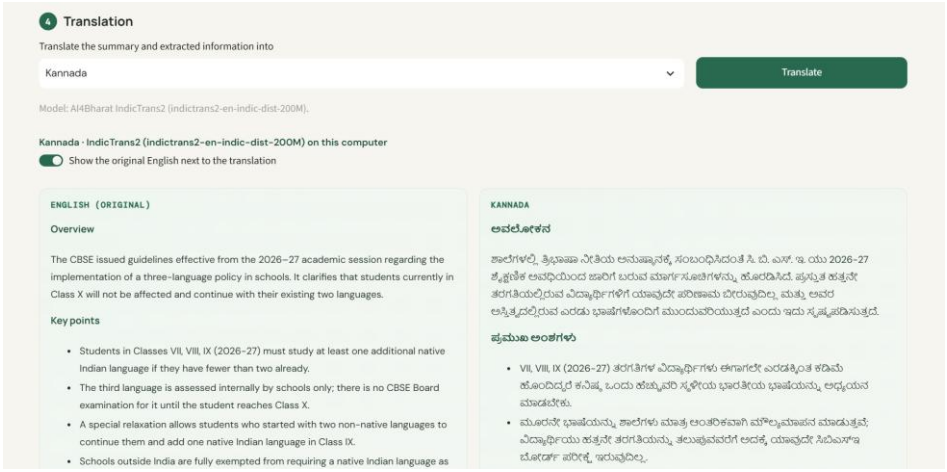


Figure 5: Translation with IndicTrans2, shown next to the original English.

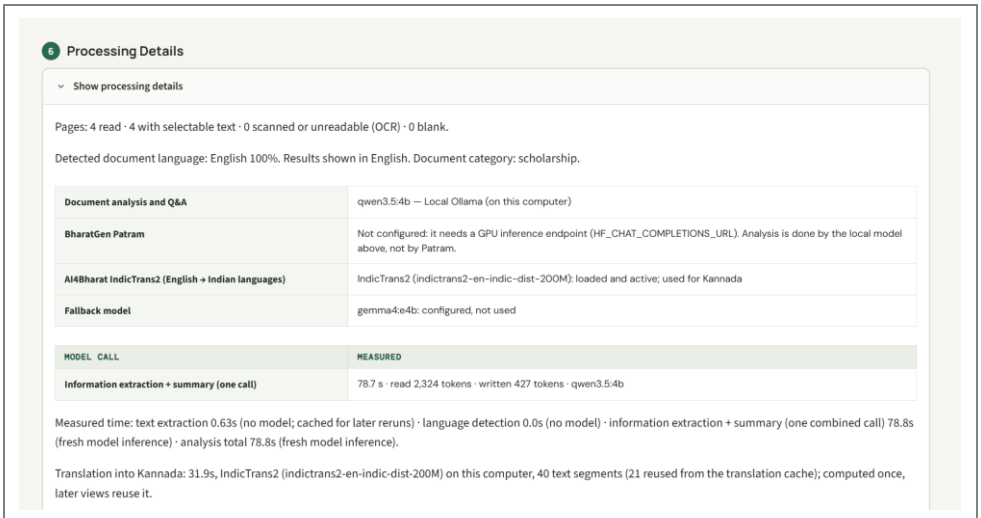


Figure 6: Processing details: the models that actually ran (Patram shown as not configured) and the time each step took.

3.3 Features

- **Inputs:** searchable and scanned PDFs of any length, Word (.docx) files and images (JPG, PNG, WEBP).
- **Results:** Document Overview, Extracted Information, Summary, Translation, Ask Questions and Processing Details.
- **Languages:** results in English, Kannada, Hindi, Tamil, Telugu or Malayalam. In Automatic mode each answer comes back in the language the question was asked in.
- **Downloads:** all results as a PDF (with Indian-script fonts), the summary as TXT or PDF, and the fields as CSV.
- **Privacy:** uploads are never written to disk; results are kept in memory for up to 8 documents and cleared when the application stops.

4. GenAI, RAG and Agentic Integration

In short: The LLM is used in three ways: to fill in a fixed form of fields and a summary, to answer questions from retrieved parts of the document (RAG), and inside a small check-and-fix loop that makes sure each answer is in the right language. Code then checks the model's output so it cannot invent numbers, dates or page references.

4.1 Structured extraction: the model fills in a form

For a document that fits in the model's context, one request returns both the fields and the summary. Ollama forces the reply into a fixed JSON structure, so it always has the expected parts. If one part comes back empty, only that part is requested again.

Long documents (for example the 127-page MH-CET brochure or the 83-page Tamil policy) do not fit. For them, the application first picks the passages most likely to contain each field (title, dates, fees, eligibility, documents, procedure, always including page 1), extracts the fields from those passages, and builds the summary from up to five section summaries.

4.2 Question answering with retrieval (RAG)

1. A short document is given to the model whole.
2. A long document is cut into overlapping chunks, each tagged with its page number.
3. The chunks that best match the question's words are retrieved. Indian-language words are matched on their stem, so grammatical endings do not stop a match. On the website, a question about a long Indian-language document is also translated into that language with IndicTrans2, so both wordings are searched.
4. The model answers only from the retrieved chunks and must cite pages. Pages it did not see are removed from the answer.
5. If the answer is not in the document, the application says "Not found in the uploaded document" instead of guessing.

4.3 Agentic loop: making sure the answer is in the right language

Each answer goes through a small decide–act–check loop instead of a single model call:

1. Detect the question's language (from its script, or from typical words if typed in English letters).
2. Decide the answer language (the one the user chose, or the question's own language).
3. Generate the answer.

4. Check the answer's script in code. If it is English but should be Kannada, translate it with IndicTrans2; for any other mismatch, ask the model to rewrite it.
5. Check again. If it is still wrong, show an error rather than an answer in the wrong language.

4.4 Checks against hallucination

- A field value containing a number that does not appear in the document is removed.
- A date must be printed in the document; it is never calculated (for example from "within 15 days").
- Source pages are found by searching the document text, not taken from the model. (In testing, the model cited "page 4" of a one-page document.)
- Leftover JSON text, duplicate points and notes such as "not mentioned" are removed before display.

5. Quantitative Testing and Error Analysis

In short: On 45 hand-checked test cases, PatraAI extracted 64.4% of fields correctly and answered 66.7% of questions correctly. Of the 31 wrong field and answer checks, 10 come from scanned Indian-language pages that OCR read badly and 10 from long documents where the right page never reached the model. When the right text reached the model, its answers were usually correct.

5.1 How the testing was done

1. Each test document goes through the same pipeline as the website: text extraction, OCR of scanned pages, field extraction, summary and question answering (evaluation/benchmark.py).
2. The predicted field and answer are compared with the hand-checked correct value (evaluation/scoring.py).
3. Small differences in writing are accepted; different facts are not. For example, "₹1,300/-" and "Rs. 1300" match, but "26.05.2026" and "25.05.2026" do not. A long answer may be worded differently if it contains the same key words and the same numbers.
4. "Not found" counts as wrong, because every answer is in its document.
5. Whether the answer cited the correct page is scored separately.

Every case was run once with qwen3.5:4b on the laptop CPU (AMD Ryzen 7 7735U, 15 GB RAM, no GPU). The first attempt was interrupted when the laptop went to sleep, and the interrupted cases were run again; between the two runs only the tidying of summary bullet points changed, which does not affect any score. All numbers in this chapter come from these runs; nothing is estimated. The full results are in evaluation/results/ in the repository.

One difference from the website: the evaluation calls the question-answering step directly. For a long Hindi, Kannada or Tamil document, the website also searches with the question translated into that language by IndicTrans2; the evaluation searched with the English question only. QA results for those documents may therefore understate the website (see 5.4).

5.2 Overall results

Table 7: Overall results

Metric	Result
Test cases evaluated	45 of 45
Field-extraction accuracy	64.4% (29 of 45 fields)
QA accuracy	66.7% (30 of 45 answers)
Answer cited the correct page	66.7%
Cases with field and answer both correct	22 of 45
Processing errors (crashes or time-outs)	0

5.3 Results by language and by format

Table 8: Results by document language

Document language	Cases	Field accuracy	QA accuracy
English	25	76.0%	88.0%
English + Punjabi	2	100.0%	100.0%
Hindi	7	57.1%	71.4%
Kannada	6	33.3%	16.7%
Malayalam	2	50.0%	0.0%
Tamil	3	33.3%	0.0%

Table 9: Results by document format

Document format	Cases	Field accuracy	QA accuracy
Image (JPG)	1	100.0%	100.0%
Legacy-font PDF (needs OCR)	3	66.7%	33.3%
Long PDF (83 pages, mostly searchable)	3	33.3%	0.0%
Mixed PDF (searchable + scanned page)	2	100.0%	100.0%
Mixed PDF (searchable + scanned pages)	6	66.7%	66.7%
Scanned PDF	7	42.9%	28.6%
Scanned PDF with OCR text layer	3	66.7%	100.0%
Searchable PDF	18	72.2%	83.3%
Word (DOCX)	2	50.0%	100.0%

English documents scored best (field 76.0%, QA 88.0% over 25 cases). Kannada (field 33.3%, QA 16.7%) and Tamil (field 33.3%, QA 0.0%) scored lowest: every Kannada document had

to be read by OCR, and the Tamil policy is 83 pages long. The Tamil and Malayalam figures rest on only 3 and 2 cases.

5.4 Failure analysis

Every wrong result was first given a category by the scorer (first table below). Each one was then traced to its root cause by checking the text PatraAI had read from the page: whether the correct value was in that text, whether the right page reached the model, and what the model did with it (second table below).

Table 10**:** Failure categories assigned by the scorer

Failure category	Wrong checks	What it means
Wrong or missing page reference	15	The answer did not cite the page that holds the answer (usually because the answer itself was wrong or "not found").
Incorrect answer	9	The answer is wrong, or it says the information is not in the document when it is.
Incorrect extracted value	9	A value was extracted, but it is not the expected one.
Missing extracted field	6	The field was left empty although the document states it.
Answer not found in document (model abstained)	4	The model said the document does not contain the answer, although it does.
OCR or document-reading failure	3	The page is scanned and almost no text could be read from it.

Table 11**:** Root causes of the wrong field and answer checks

Root cause	Wrong checks	Example cases
Right page not found in a long document	10	brochure-01, brochure-02, brochure-03, chd-02, chd-04, tapolicy-01, tapolicy-02, tapolicy-03
Scanned page read badly (OCR)	10	fln-01, fln-02, fln-03, bagless-02, bagless-03, klgazette-02
Another number taken as the reference number	4	chd-03, apscheme-01, klgazette-01
Model missed a value that was in the text	4	chd-04, tnscholar-02, odisha-02, docx-02
A check or the scorer was too strict	3	law-02, tnscholar-01, apscheme-03

The 31 wrong field and answer checks have five root causes. Page-reference errors are not counted here: 15 answers cited the wrong page or none, but only 3 of them had a correct answer; the rest are a side effect of a wrong or "not found" answer.

- **Right page not found in a long document.** In a long document only selected passages reach the model, and both the field selector and the question search work by matching words. In the 127-page MH-CET brochure the issuer (page 1), the fee (page 75) and the income limit (page 17) were not selected. In the 83-page Tamil policy the field selector, whose keywords are English, picked only page 1, and the English questions could not match Tamil text, so the right pages never reached the model and it correctly said it could not find the answer. In the Hindi circular the signatory's page (page 4) was not passed to the model; page 1 contains a garbled copy of the name ("संगीता जेहा"), which the model read as "Sangita Jha".
- **Scanned page read badly (OCR).** On scanned Kannada and Malayalam pages, the 4-billion-parameter vision model read numbers well but garbled most words (the Kannada reference number came out as "E-8146 DSERT/EVG/ ಸಾಧುವ ತೀವರತ/2026-27": digits right, words wrong). On page 1 of the Kannada FLN circular, the page with the department, reference number and date, it read nothing at all, and the backup model also returned nothing, so every check on that document failed. When a value is missing from the text that was read, no later step can recover it.
- **Another number taken as the reference number.** Government orders quote many other orders, letters and registration numbers. The model sometimes returned one of these instead of the document's own number: orders cited inside the AP scheme order (G.O.Ms.No.9) instead of its own G.O.Ms.No.26, the Kerala Gazette's postal registration number (KERBIL/2012/45073) instead of G.O.(P) No. 145/2013, and another code printed on the Hindi circular (CBC2152/11/0002/2526) instead of 2-5/2020.
- **Model missed a value that was in the text.** Sometimes the right text reached the model and it still left the value out: the required documents in the Hindi circular and in the Odisha form were left empty, and for the CS3235 project sheet the model listed other instructions but not the requirement of 30–50 test cases. In one case a shortcut was to blame: questions that mention an amount are first answered by quoting an amount matched by a fixed text pattern, and for the Tamil Nadu scholarship letter the pattern matched "Rs 30" on page 12 instead of the Rs 2.00 lakh maximum on page 1.

- **A check or the scorer was too strict.** The date check removed a correct date because it only understands dates written in digits or with English month names: the Hindi circular's "24 अक्टूबर, 2025" was dropped. The automatic scorer also rejected two right results: "Class I to Class XII" did not match the expected "Classes I to XII", and the Tamil Nadu letter's reference number was extracted correctly but with another order's number added. True accuracy is therefore slightly higher than reported.

Examples: expected versus predicted

Table 12**:** Examples of wrong results

Case	Check	Expected (correct)	PatraAI predicted	Root cause
fln-01	issuing department	Department of State Educational Research and Training | DSERT | ರಾಜ್ಯ ಶೈಕ್ಷಣಿಕ ಸಂಶೋಧನೆ ಮತ್ತು ತರಬೇತಿ ಇಲಾಖೆ	Government of Karnataka, Department of School Education and Literacy	OCR
bagless-02	reference number	E- 8146 DSERT/EVG/ಸಂಭ್ರಮ ಶನಿವಾರ/2026-27 | E-8146 DSERT/EVG/Sambhrama Shanivara/2026-27	E-8146 DSERT/EVG/ ಸಾಧುವ ತೀವರತ/2026-27	OCR
klgazette-02	answer	Right to Service Act | സേവനാവകാശ	The document does not explicitly state under which specific Act (e.g., Indian Marriage Laws) the notificati...	OCR
brochure-02	amount	₹ 1300/- | Rs. 1300	(nothing)	Page not found
tapolicy-01	answer	58,800 | 58800	The document does not contain information about the total number of schools in the Tamil Nadu School Educat...	Page not found
chd-04	answer	Sangita James | संगीता जेम्स	The vacancy circular was signed by Sangita Jha, who is identified in the document as an Administrative Offi...	Page not found
apscheme-01	reference number	G.O.MS.No. 26 | G.O.Ms.No.26	G.O.Rt.No. 326, G.O.Ms.No.9, Comp. No. 2858012	Wrong number picked
klgazette-01	reference number	G.O.(P) No. 145/2013/G.Edn | 145/2013	KERBIL/2012/45073 / KL/TV(N)/634/2012-14	Wrong number picked
odisha-02	required documents	Voter ID | Aadhaar | IT returns | RoR	(nothing)	Model missed it
tnscholar-02	answer	Rs.2.00 Lakh | 2.00 Lakh | 2 lakh	The document states: RS 30. [Page 12]	Model missed it
law-02	date	24 अक्टूबर, 2025 | 24 October 2025	(nothing)	Too strict
apscheme-03	answer	Classes I to XII | Class I to XII | I to XII	The "THALLIKI VANDANAM" scheme covers students from Class I to Class XII, including Intermediate Education....	Too strict

Every check, right or wrong, is listed in evaluation/results/evaluation-results.csv in the repository.

5.5 Speed before and after optimisation

The same five documents were timed before and after the speed work, with the same models and the same timing script. Accuracy on these documents stayed the same.

Table 13**:** Speed before and after optimisation

Document	Analysis time	Translation time (Kannada + Hindi)	Fields correct
Telangana government order (1 page)	76 s → 58 s	43 s → 32 s	2/2 → 2/2
CBSE press release (4 pages)	107 s → 84 s	59 s → 36 s	3/3 → 3/3
CS3235 project sheet (Word)	92 s → 78 s	121 s → 40 s	1/2 → 1/2
AP scheme order (7 pages)	195 s → 148 s	104 s → 62 s	2/3 → 2/3
Hindi circular (7 pages)	149 s → 109 s	53 s → 21 s	2/3 → 2/3
Total	618 s → 476 s (23% less time)	380 s → 191 s (50% less time)	unchanged

What made it faster:

- one combined model call for fields and summary, with shorter instructions and a shorter reply;
- the model starts reading the prompt while the user is still looking at the preview;
- every step is cached, so a language change or a repeated question does no new work;
- IndicTrans2 translates sentences in batches of 32. On 98 test sentences, middle-sized batches were fastest (74 s with 24, 71 s with 48), while very small and very large batches were slower (109 s with 8, 113 s with 96).

5.6 Automated tests

98 automated tests (pytest) check the application without needing the AI model: page and OCR handling, cleaning of model output, language detection, translation, the scoring rules, the launcher, the PDF download, and full runs of the web page with stand-in model services. All 98 pass.

Table 14**:** Automated tests by file

Test file	Tests
test_app_e2e.py	5
test_evaluation.py	8
test_indictrans2.py	6
test_launcher.py	1
test_multilingual_qa.py	18
test_pdf_pages.py	31
test_results_pdf.py	1
test_summary_and_cleanup.py	20
test_translation.py	8

6. Demo, Documentation and Repository

In short: All code, the test dataset, the results and this report are in the public GitHub repository. The application starts with a double-click.

6.1 GitHub repository

Repository: https://github.com/pavankumarkn04/PatraAI

Table 15**:** Contents of the repository

Path	What it contains
README.md	Setup, models (including the Patram decision), how the pipeline works, results and limitations
Indian.py	The Streamlit web application
document_processing.py	Text extraction, OCR, model requests, checks and retrieval
question_answering.py, language_detection.py	Multilingual question answering and language checks
translation.py, indictrans2_processor.py, localization.py	IndicTrans2 translation
document_profiles.py	Document categories, fields shown and suggested questions
run_app.py, Start/Stop PatraAI.bat	One-click launcher
evaluation/dataset/	The 45 test cases, the 18 documents and how each answer was verified
evaluation/results/	Evaluation report, every check (CSV) and every prediction (JSON)
tests/	98 automated tests
docs/	This report and the application screenshots

The Hugging Face token (.env) and caches are excluded from the repository by .gitignore.

6.2 How to run

1. Install Python 3.11 and Ollama. Accept the IndicTrans2 terms on Hugging Face and put a read token in a .env file (HF_TOKEN=...).
2. Double-click Start PatraAI.bat. It installs missing packages, starts Ollama, downloads the models the first time, and opens the application in the browser.
3. Double-click Stop PatraAI.bat to stop it.

To repeat the evaluation: py -3.11 evaluation/run_evaluation.py. To run the tests: py -3.11 -m pytest -q.

6.3 Demo script (about 10 minutes)

1. Upload the CBSE press release (searchable PDF) and show the page preview.
2. Click Analyze. Show the extracted information with source pages, and "Not found in document" for missing fields. Show the summary.
3. Switch the response language to Kannada. The results appear in Kannada without analysing again.
4. Ask in Kannada "ಈ ದಾಖಲೆಯನ್ನು ಯಾರು ಹೊರಡಿಸಿದ್ದಾರೆ?" (Who issued this document?) and then in Hindi. Show the answer language and the source page.
5. Ask something the document does not say. The answer is "Not found in the uploaded document".
6. Upload a scanned page (the JPG letter). OCR runs only for that page.
7. Open Processing Details: the models that actually ran and how long each step took.
8. Download all results as a PDF.

7. Limitations and Future Work

- **Patram was not run.** It needs a GPU server. With one, a single setting sends all analysis to Patram. Comparing Patram with qwen3.5:4b on the same 45 cases is the most important next step.
- **OCR of Indian scripts.** The 4B vision model garbles Kannada and Malayalam words on scanned pages. PatraAI already uses Tesseract with Indian-language packs automatically when it is installed; it was not installed for this evaluation, and testing it is the first next step.
- **Long documents.** Passages are chosen by matching words, and the field selector's keywords are English, so long Indian-language documents get poor coverage. Multilingual retrieval (for example with embeddings) would address this.
- **Evaluation question path.** The evaluation searched long Hindi, Kannada and Tamil documents with the English question only, while the website also searches with an IndicTrans2 translation of the question. Re-running those 16 cases through the website's own question function is future work.
- **Telugu documents are not tested.** Tamil (3 cases) and Malayalam (2 cases) also have few cases, so their accuracy figures are rough.
- **Slow on a CPU.** One analysis takes about 1–2.5 minutes, plus one extra model call per scanned page. A GPU would make it interactive.
- **Automatic scoring.** A correct answer worded very differently from the expected one can be counted as wrong. Summaries and translations are not scored.

8. Conclusion

PatraAI is a working, general-purpose Indian document understanding tool. It reads searchable, scanned, legacy-font and Word documents; extracts details with their source pages; writes a short summary; translates the results with IndicTrans2 into five Indian languages; and answers questions in six languages. On 45 hand-checked test cases from 18 real public documents it reached 64.4% field-extraction accuracy and 66.7% QA accuracy. Most errors trace back to OCR of scanned Indian-language pages and to word-matching retrieval in long documents, which point to the two clearest next steps: better OCR and multilingual retrieval. After optimisation, analysis takes 23% less time and translation 50% less time, with the same accuracy. The specified Patram model could not run on the available laptop; the application supports it through a server setting and always reports which model actually answered.

Code, data and results: https://github.com/pavankumarkn04/PatraAI

References

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[3] J. Gala et al. IndicTrans2: Towards High-Quality and Accessible Machine Translation Models for all 22 Scheduled Languages of India. Transactions on Machine Learning Research, 2023.

[4] Qwen team. Qwen3.5 models, served with Ollama. https://ollama.com

[5] Streamlit, PyMuPDF, python-docx and Hugging Face Transformers documentation.

[6] PatraAI project repository. https://github.com/pavankumarkn04/PatraAI

Appendix A. All Test Cases and Results

Field: whether the expected field was extracted correctly. Answer: whether the question was answered correctly. Page: whether the answer cited the correct page.

Table 16**:** All test cases and results

Case	Language	Question	Field	Answer	Page
brochure-01	English	Who issued this information brochure?	Yes	No	No
brochure-02	English	What is the application fee for General Category candidates from Maharashtra State?	No	Yes	Yes
brochure-03	English	What is the parental annual income limit for the reservation scheme for Maharashtra State candidates?	No	Yes	Yes
cbse-01	English	What is the date of the press release?	Yes	Yes	Yes
cbse-02	English	Out of the three languages, how many must be Bhartiya Bhashas?	Yes	Yes	Yes
cbse-03	English	In how many scheduled Bhartiya Bhashas are dedicated R3 textbooks for Class VI being made available?	Yes	Yes	No
chd-01	Hindi	Which office issued this circular?	Yes	Yes	Yes
chd-02	Hindi	How many posts of Assistant Director (Language) are to be filled?	Yes	No	No
chd-03	Hindi	What is the maximum age limit for appointment on deputation?	No	Yes	Yes
chd-04	Hindi	Who signed the vacancy circular?	No	No	No
law-01	Hindi	Which department issued this circular?	Yes	Yes	Yes
law-02	Hindi	Which post is to be filled on deputation?	No	Yes	Yes
law-03	Hindi	What is the pay matrix range for the post?	Yes	Yes	Yes
fln-01	Kannada	Which department issued this circular?	No	No	No
fln-02	Kannada	For how many days was the bridge programme conducted in June?	No	No	No
fln-03	Kannada	On which Saturday of every month should the head teachers' meeting on FLN be held?	No	No	No
bagless-01	Kannada	What is the date of this circular?	Yes	Yes	Yes
bagless-02	Kannada	How many self-learning activity modules were prepared for students?	No	No	No
bagless-03	Kannada	Which module is scheduled for January 2027?	Yes	No	Yes
tnscholar-01	English	What is the parental annual income limit for this scholarship?	No	Yes	Yes
tnscholar-02	English	What is the maximum scholarship amount per annum?	Yes	No	No
tnscholar-03	English	To whom should applications of Backward Classes (BC) students be sent?	Yes	Yes	No
tnscholar-img-01	English	What is the letter number?	Yes	Yes	Yes
apscheme-01	English	How much financial assistance is given per child per annum under the scheme?	No	Yes	Yes
apscheme-02	English	How much is deducted at source per student?	Yes	Yes	Yes
apscheme-03	English	Which classes are covered by the scheme?	Yes	No	Yes

Case	Language	Question	Field	Answer	Page
tapolicy-01	Tamil	How many schools did the School Education Department have as of March 2024?	No	No	No
tapolicy-02	Tamil	How many students were there as of March 2024?	Yes	No	No
tapolicy-03	Tamil	What share of total state expenditure is allocated to school education?	No	No	No
klgazette-01	Malayalam	What is the G.O. number of this notification?	No	No	Yes
klgazette-02	Malayalam	Under which Act is this notification issued?	Yes	No	No
tgschedule-01	English	What is the last date for online application submission and fee payment?	Yes	Yes	Yes
tgschedule-02	English	When is the certificate verification of eligible candidates?	Yes	Yes	Yes
tgorder-01	English	What amount is administratively sanctioned in this order?	Yes	Yes	Yes
tgorder-02	English	Who signed this government order?	Yes	Yes	Yes
tgproceedings-01	English	What honorarium does the District Collector receive as chairperson of the district committee?	Yes	Yes	Yes
tgproceedings-02	English	On which dates were certificate verification and interviews originally scheduled for all institutions?	Yes	Yes	Yes
obc-01	English	Who can issue this certificate?	Yes	Yes	Yes
obc-02	English	Which website gives the latest guidelines on the Central List of OBCs?	Yes	Yes	Yes
odisha-01	English	What is the gross annual family income limit for the EWS certificate?	Yes	Yes	Yes
odisha-02	English	Above what size of agricultural land is a family excluded?	No	Yes	No
punjab-01	English + Punjabi	What is the RTS service number of this form?	Yes	Yes	Yes
punjab-02	English + Punjabi	When was this form last revised?	Yes	Yes	Yes
docx-01	English	Which BharatGen model is suggested for the Indian Document Understanding Tool?	Yes	Yes	Yes
docx-02	English	How many representative test cases are required?	No	Yes	Yes